H. G. Kolsky

COMPUTER ORGANIZATION AND SYSTEMS

Lecture Notes

Lecture 1: Monday, April 9, 1962

I. Introduction

A. General System Concepts - Ref: Goode and Machol

System - Method or order, organization, classification, codification

- A complete assemblage arranged in essential, rational dependence or connection
- an organized whole

Steps in designing and implementing a system:

Phase (1) Problem Definition

- solidify objectives
- analyze present methods
- evaluate improvements or expansions
- establish flow and methods of processing, methods to be used to select equipment
- Write a statement detailing the requirements, objectives, costs and installation schedule

Phase (2) Design Approach

- establish a basis for creating design configurations
- evaluate each configuration vs. system requirements and availability of required components
- for the best designs; estimate time, cost, personnel and "pay off"
- Models or simulation of system to estimate the effect of changes in design parameters
- documentation

Phase (3) Detailed Design

- detailed descriptions are written -- Engineering "Specs.", Program "Specs."
- equipment is carefully evaluated vs. each job
- analysis of theory, methods, techniques, computer programming, etc., to find gaps in design
- analysis of human factors : selection, training, personnel policies, man-machine relations, error correction procedures, etc.
- reliability planning -- equipment maintenance, human training, balance and tolerances, reporting procedures
- transition schedule from present to new system

Phase (4) Implementation

- equipment procurement, installation, tested
- conversion of input to forms needed by system
- write and check computer programs
- testing methodology on controlled samples
- training courses and O. J. T.
- acceptance tests
- (acceptance date)
- follow up -- retrofits -- plans, etc., generates new Phase I's

Some Examples of Large Systems

Blind landing systems - GCA

Reservisor system

SABRE

Denver Traffic - control system

Telephone message accounting

Some Examples of Large Systems (continued)

Air Material Command Logistics

Automatic Post Office

SAGE

Radiosonde system

Weapons systems, e.g., NIKE, BOMARC

Insurance Company records

Stock brokers

Some Properties of Systems

- a system has "integrity"
  may or may not be centrally controlled

- large systems - large number of parts, types of parts, numbers of functions, large in cost

"Complex" - a change in one variable will affect many other variables rarely in a linear way

"Automatic" - some degree

"Statistical" - distributed loads

"Competitive" - when some rational agent is trying to destroy it

"The Team Approach" - 5-12 people

A system is built of: procedures, tools, parts

Constraints on Systems

- "interface conditions" with other systems
- data flow rates and volumes
- men available at training levels
- money available
- time available

Constraints on Systems (continued)

- tools available vs. time
- parts available vs. time

Some of Pitfalls in Systems Design (mostly stem from lack of integrity)

(1) because of competitive situation -- take on jobs which are not possible technically

(2) "get contract and then negotiate" attitude --- stall -- use time arguments

(3) get job with people who will not be put on job -- lack of knowledge

(4) doing good Phase I, but no follow up

(5) overemphasizing some economic or technical aspect at expense of whole

(6) goals of systems developers different from time job -- or "wander off path"

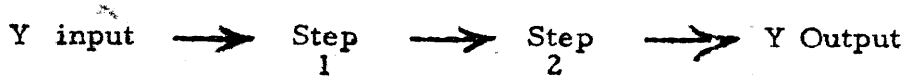
(7) trying to go through phases too fast --- "profit is in hardware" attitude

Problem 1

Problem in Systems Design

Design system to process "product Y" under various constraints of rate, cost, and machine component capacity:

A two-step process:



Machine Possibilities:

Name	Task	Rate of Processing	Cost of Machine
A_1	Step 1	3 Y's/day	$10
A_2	Step 1	10 Y's/day	21
B_1	Step 2	2 Y's/day	15
B_2	Step 2	6 Y's/day	33

Assume any number of A's and B's can be connected.

To do: (1) Plot cost vs. number Y's processed up to 18 Y's per day

(2) Plot cost per Y vs. number of Y's processed

(3) If the flow of Y's varies from 8 to 15 per day which System would you propose?

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Lecture 2: Wednesday, April 11, 1962

B. History of Computers

Refs.
- F. L. Alt (1958)
- M. V. Wilkes (1956)
- B. V. Bowden (1953)
- Ned Chapin (1955)
- Proc. ACM (1952)

Did arithmetic give birth to mechanical computing devices or vice-versa?

Addition	Roman Numerals		Multiplication
525	DXXV	325	CCCXXV
+ 131	CXXXI	13	XIII
656	DCLVI	975	CCCXXV
		325	CCCXXV
		4225	CCCXXV
			CMLXXV
			MMMCCL
			MMMMCCXXV

(Read from Page 1 ACM)

Abacus - explain operation

Hindu-Arabic numbers, "The missing column"

Dates (Using Table from p. 232 ff Chapin)

1580	Napier and Briggs - logarithm
1632	Oughtred - slide rule
1642	Pascal
1671	Leibnitz - first advocated binary number system
1770	Hahn - first dependable 4-process calculator
1820	Thomas - crank machine "Thomas-type" machine
1875	Baldwin "Baldwin-type"
	(Monroe joined with Baldwin in 1911)
	Friden
	Marchant - Avery
1812	Babbage (1792 - 1871)

Differences:

M	M^3+M^2	D_1	D_2	D_3
0	0			
1	2	2	8	6
2	12	10	14	6
3	36	24	20	
4	80	44	26	6
5	150	70	32	6

Babbage's Life

1822 - Small working model of difference engine
1833 - Royal Society support
1842 - Official support withdrawn and project abandoned
1862 - Model was exhibited at Internation Exposition
1859-63 - Small models were used by insurance companies

1833 - "Analytical Engine"
- Parallel Carriers 20 digit No. (67 bits)
- 20 decimals and 6 order differences
- To store 1, 000 50-digit numbers
- Punched Cards
- Looping Control
- Conditional Branching

(Read p. 11-12 of "Faster Than Thought"
12-13
p. 17

1890 - Hollerith

Modern Computers: Family Tree

Differential Analyzers: Fire Control Devices

1944 - Mark I "Roots"
1948 - SSEC
Bell Relay Computer
1947 - Eniac
1949 - EDSAC "Trunk"
1950 - SEAC
1949 - CPC

(Read ACM Proc., page 31)

(Read p. 236 "Intro. to Automatic Computers" - Chapin)

Family Tree: "Branches"

IAS type - parallel
Edvac - serial
Military - (speed and reliability)

Mention IBM's late entry into large machine area

- follow various "branches"
- stress importance of CPC and 701
- mention 702 - UMVAC battle

Problem 2

Try dividing CCCXXV by XIII

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Lecture Notes

Lecture 3 April 13, 1962

1. Discuss Roman numeral division problem - stress use of multiples of X, V.

2. Finish discussion of history & IBM machines.

   Stress importance of CPC - "The Model-T of Computing"

   Three main branches of Tree:

(1)	Commercial	(serial - decimel)	702, 705, ---- 7074
(2)	Scientific	(parallel - binary)	701, 704, ---- 7030, 7094
(3)	Military	(reliability - rugged)	FSQ7 ------- FSQ32

   What will next "off shoots" on the tree be like?
   - multiprocessing
   - process control
   - "real time - on line"
   - teleprocessing
   - small individualized computers
   - giant centralized computers

3. Show movie F-15, 3 "The Information Machine"*

   Problem 3: Take notes and write 1-page summary of this movie.

4. Start next section.

II. Data Representation

   A. General discussion: What is a message? What is meaning? What is data?

      Example: Now is the time for all good men to come to the aid of their country.

      Jetz, ist der Zeit für alle gute Männer seinem Vaterland zu Hilfe kommen.

      -. --- .-- .. .... -..... -..--., etc.

* Note: Next time this shall be at end of period, not start.

Discuss meaning of the following terms using above examples:
- message
- information
- data
- meaning (translation vs coding context)
- character sets
- intelligence
- letter
- symbol
- bit
- redundancy - efficiency

Spoken English is about 50% redundant - i. e. half of sounds can be reconstructed from context.

Written English is about another 50% redundant -- mention efficiency of shorthand ... context ...



nau sth tm f a g mn etc.

B. Number Bases (start only) - review meaning of positional number systems

$$N = \cdots a_2 r^2 + a_1 r^1 + a_0 + a_{-1} r^{-1} + \cdots$$

Problem A:

Convert $(433)_{10}$ to base 8, 5, 3, 2 by any method you can find.

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Lecture Notes

Lecture 4: Monday, April 16, 1962

B. Number Bases and Representation

Ref. P. Siegel, Chap. 2.

1. Explain meaning of positional number system.

$$433 = 4 \times 10^2 + 3 \times 10^1 + 3 \times 10^0$$

In general:

$$\ldots a_3\ a_1\ a_0 .\ a_{-1}\ a_{-2} \ldots = \ldots + a_2 r^2 + a_1 r_0^2 + a_0 r^0 + a_{-1} r^{-1} + a_{-2} r^{-2} + \ldots$$

2. Number of digits n required to represent a number N with base b:

$$N = b^n$$

for 2 bases: $N = b_1^{n_1} = b_2^{n_2}$

$$M_1 \log_{10} b_1 = M_2 \log_{10} b_2$$

$$\frac{M_2}{M_1} = \frac{\log_{10} b_1}{\log_{10} b_2}$$

$\log_{10} 10 = 1.0$

$\log_{10} 2 = 0.30103$

$\log_{10} 3 = 0.47712$

So binary takes $\frac{M_2}{M_1} = 3.321 \approx 3\ 1/3$

Remember $2^{10} \approx 10^3$ $\quad (2^{10} = 1024.)$

3. Examples of different positional number systems

Decimal 10	Hexa-Decimal 16	Duo-Decimal 12	Octal 8	Quarternary 4	Ternary 3	Binary 2
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	10
3	3	3	3	3	10	10
4	4	4	4	10	11	100
5	5	5	5	11	12	101
6	6	6	6	12	20	110
7	7	7	7	13	21	111
8	8	8	10	20	22	1000
9	9	9	11	21	100	1001
10	A	A	12	22	101	1010
11	B	B	13	23	102	1011
12	C	10	14	30	110	1100
13	D	11	15	31	111	1101
14	E	12	16	32	112	1110
15	F	13	17	33	120	1111
16	10	14	20	100	121	10000
17	11	15	21	101	122	10001
18	12	16	22	102	200	10010
19	13	17	23	103	201	10011
20	14	18	24	110	202	10100
40	28	34	50	220	1111	101000
100	64	84	144	1210	10201	1100100

Point out "family resemblance" between 16, 8, 4, 2.

Also, mention +, 0, - system (signed Ternary)

931			
0			0
1			+
2		+	-
3		+	0
4		+	+
5	+	-	-
6	+	-	0
7	+	-	+
8	+	0	-

4. What are the advantages of binary?

1. Binary 1 and 0 can represent the "yes" and "no" of formal logic.

2. Binary numbers are easy to mechanize (switches, flip-flops, etc.,),

3. Binary arithmetic is very simple.

Truth tables:

Addition:

Decimal

Addend \ Augend	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	10
2	2	3	4	5	6	7	8	9	10	11
3										
4										
5										
6					etc.					
7										
8										
9										

45 carries out of 100 cases

$P_c = 0.45$

Binary

Addend \ Augend	0	1
0	0	1
1	1	10

Comparative Binary case

$P_c = (3.321)(.25) = 0.83$

Multiplication: Binary

Multiplier \ Multiplicand	0	1
0	0	0
1	0	1

5. Conversion from one base to another.

1st METHOD: TABLE LOOK-UP.

Binary	=Decimal
.0001	.0625
.001	.125
.01	.25
.1	.5
1.	1.
10.	2
100.	4
1000.	8
10000.	16
100000.	32
1000000.	64
10000000.	128
100000000.	256

Decimal	Binary
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
20	10100
30	11110
40	101000
50	110010
60	111100

Example of decimal to binary conversion: $(191)_{10} = (\ ?\ )_2$

191	
-128	10000000
63	
-32	10100000
31	
-16	10110000
15	
-8	10111000
7	
-4	10111100
3	
-2	10111110
1	
-1	10111111
0	

Example of binary to decimal $(111011)_2 = (\ ?\ )_{10}$

111011	
-110010	50
1001	
-1001	59
0	

Notice arithmetic in both above is in "converted-from" base.
We can do it in "converted-to" base by using opposite table.

2nd METHOD: DIVIDING (from Siegel - Page 36)
SUCCESSIVELY BY THE BASE, KEEPING REMAINDER.

2)137			
2)68	Q = 68	R = 1	LSD
2)34	34	0	
2)17	17	0	
2)8	8	1	
2)4	4	0	
2)2	2	0	
2)1	1	0	
	0	1	MSD

$N = a_2 b^2 + a_1 b^1 + a_o$

$N = b(a_2 b + a_1) + a_o$

Ans. = 10001001

3rd METHOD: MULTIPLYING SUCCESSIVELY BY THE BASE FOR FRACTIONAL NUMBERS.

	Product	Carry	
0.413 x 2	0.826	0	MSD
0.826 x 2	1.652	1	
0.652 x 2	1.304	1	
0.304 x 2	0.608	0	
0.608 x 2	1.216	1	
etc.			Ans. 0.01101...

4th METHOD: USING MEANING OF BINARY ZERO (Table Look-up, plus shifting and adding)

$(855)_{10} = (?)_2$

If we knew	5	=	101
	10	=	1010
	40	=	101000
	160	=	10100000
	640	=	1010000000

Add:	5	101	
	10	1010	
	40	101000	
	160	10100000	
	640	1010000000	
	855	1101010111	= $(1527)_8$

5th METHOD: USING RULES OF NUMBER FORMATION AND SMALL TABLE.

$$(372)_{10} = 3 \times 100 + 7 \times 10 + 2$$

2	=	10
3	=	11
7	=	111
10	=	1010
100	=	1100100

$$372 = (11)(1100100) + (111)(1010) + (10) = (3)(144) + (7)(12) + (2)$$

$$= 100101100 + 1000110 + 10 = 454 + 106 + 2$$

$$= 101110100. = 564$$

6. Coded Decimal numbers:

Coding each decimal in 4 bits:

52963 = 0101 0010 1001 0110 0011

Also called a "weighted bit code".

8	4	2	1	
0	1	0	1	= 5

(See Siegel, page 46, for examples of other codes.)

(Table 11) 2421 8421 5421 5211 (4311 3321)

"Excess-3" code "Excess-6" code

Reasons for some codes:

(1) self-checking - redundancy - forbidden combinations

(2) self-complementing:

(Table 12) 2421 5211 4311 4221 3321

"Excess-3" code.

Notice that "normal" code 8421 is not self-complementing.

Alpha-numeric codes.

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Lecture Notes

Lecture 5 Wednesday, April 18, 1962

II. Data Representation (cont'd)

Base b system requires b distinct symbols.
Give examples: Base 2, 5, 8, 16, etc.

When these are not available, they can individually be encoded as combinations of other symbols.

Examples: Decimal coded lexical

61	A
62	B
63	C
etc.	

Binary coded decimal

	604	Univac	650 drum	650 arith
0	0000	0011	00110	0100001
1	0001	0100	00011	0100010
2	0010	0101	00101	0100000
3	0011	0110	01001	0101000

Code: - a system of words or other symbols arbitrarily used to represent words. (note: arbitrarily)

Cipher - a method of secret writing that substitutes other characters for the letters intended, or transposes the letters after arranging them in blocks.

Truth table: - a statement of all the conditions of a problem in tabular form.

Quantitative measure of information given by symbols.

base b: L_1 symbols give M_1 possible messages.

$$M_1 = b^{i_1}$$

define "Information" as $i = \log_b N$ where N = no. of equi-probable messages.

So information is additive: $i_1 ti_2 = \log_b m_1 + \log_b M_2$

Recall meaning of $\binom{A}{B}$ $\quad \frac{A!}{(A-B)!\ B!}$ permutations

"A things taken B at a time"

"Efficiency" of coding system:

$$\text{efficiency} = \frac{\text{number of combinations used in system}}{\text{number of possible combinations}}$$

example: If we use 4 bits to represent decimal numbers.

$$\text{eff.} = \frac{\text{10 symbols used}}{\text{16 symbols possible}} = 0.625$$

"Redundancy" = 1 - efficiency.

So, using 4 bits for 10 symbols has a redundancy of 1 - 0.625 = 37.5%

WHY ARE THERE SO MANY DIFFERENT SYSTEMS?

Redundancy allows us to choose certain combinations as valid and others as invalid. This choice is arbitrary. Other considerations may be used to help make the choice. Some such are:

(1) Weighting
    semi-weighting
(2) Self-complementing
(3) Error-detecting
    Symmetric
(4) Arithmetic-easing

Redundancy used for error detection:

The "2 out of 5" code: - we use only those combinations which have 2 "1" bits and 3 "0"'s.

00000	01000	10000	11000 ←
00001	01001 ←	10001 ←	11001
00010	01010 ←	10010 ←	11010
00011 ←	01011	10011	11011
00100	01100 ←	10100 ←	11100
00101 ←	01101	10101	11101
00110 ←	01110	10110	11110
00111	01111	10111	11111

Only the 10 combinations marked out of the 32 possible are used.

$$\binom{5}{2} = \frac{5!}{2!\,3!} = \frac{5.4.3.2.1}{2.1.3.2.1} = 10$$

If one of the other combinations occurs, it represents an error -- a bit changed.

$$\text{efficiency} = \frac{10}{32} = 0.3125$$

$$\text{redundancy} = 0.6875$$

PARITY CHECKING: (another type of redundancy) -

	421	p
0	000	1
1	001	0
2	010	1
3	011	0
4	100	1
5	101	0
6	110	1
7	111	0

Add another bit so that:

p = 1 if no. of 1's is even
0 if no. of 1's is odd

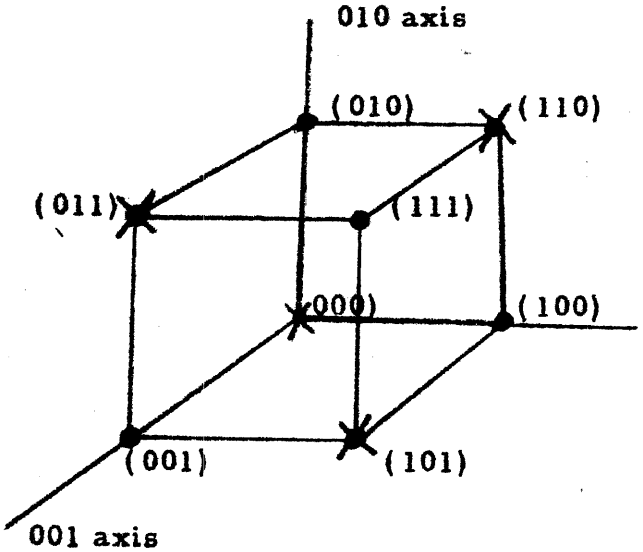
If an error in one bit (including p) occurs, the checking circuits will detect it.

We have used up 1/2 the possible codes for checking:

$$\text{efficiency} = \frac{8}{16} = \frac{1}{2}$$

redundancy = 50%

Describe HAMMINGS' CUBE: - - Three bit case



	21 p
0	00 1
1	01 0
2	10 0
3	11 1

● allowed
✕ not allowed

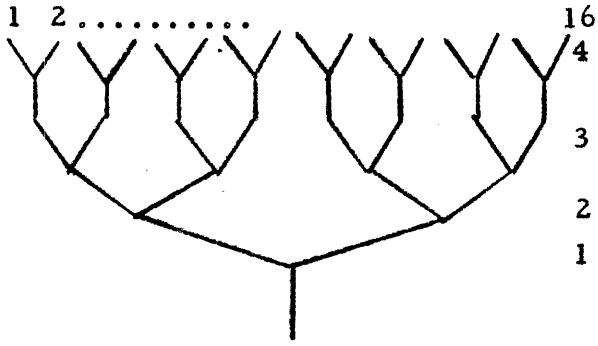
Allowed combinations (001), (010), (101), (110) are all "distance 2" apart on the cube, i. e. two bits must be in error to give another valid combination.

C. Transmission of Data

1. Information Theory or Communication Theory

- Meaning of "bit of information" --- choice between 2 alternatives.

Example: choice of 1 of 16 standard birthday messages.



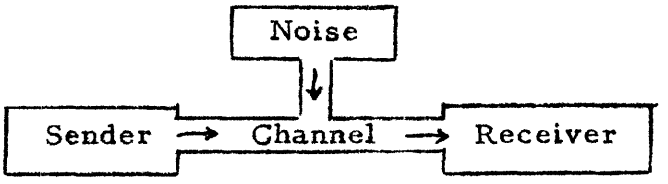
4 decisions needed to select which of 16 messages

$16 = 2^4$

$\max N = 2^n$

Maximum number of selections which can be made with n bits of information.

- Concept of "noise" in transmission of data



The behavior of the noise signal vs time can influence the whole system design.

The ratio of the signal to the noise is what counts - not absolute value.

Channel Capacity: Shannon's Max. Theoretical value.

$$C = B \log_2 \left(1 + \frac{Ps}{Pn} \right)$$

where B = band width (cps), $\frac{Ps}{Pn}$ = Signal to Noise power ratio

C = capacity (bits per sec)

Practical systems do not increase this rapidly with B, nor come close to B.

Measure of Information transmitted:

Mention: 1 bit of information per letter is about average in English -- choice is not random from letter to letter. That is: on the average each letter contributes 1 choice not 26.

Measure of information per symbol (called "Entropy")

$$I = -\sum P_n \log_2 P_n$$

Pn = probability of the nth symbol being chosen

(15 bits - how many are needed?)

Sequence of equal 1's & 0's: 0110011110010110

unequal : 111101111110101

If Po = 1/6 P_1 = 5/6 then $I = -\left[1/6 \log_2 1/6 + 5/6 \log_2 5/6\right]$

So only (15) (.651) = 10 bits are necessary $= -1/6\ (-2.59) + 5/6\ (-.263) = .651$

Problems:

(1) How many bits per second can be sent over a television channel. Bandwidth = 4 Mc. signal to noise ratio = 10,000.

(2) How many bits per second over a telephone line. Bandwidth = 1.0 KC. signal to noise = 10.

(3) If one sailor signals by flags, etc. to another when a blonde passes by the corner and when a brunette passes by -- what is information content of the message if:

$P_{blonde} = 1/64$, $P_{brunette} = 63/64$, when he sends (1) "blonde"
(2) "brunette" ?

How can the efficiency of the transmission be improved?

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Lecture Notes

Lecture 6: Monday, April 23, 1962

Boolean Algebra Ref. Phister Chap. 3
Shannon TrAIEE 1938

Basic Postulates: Class K & rules "+ or" and "· and".

Definitions:
(1) If A is a member of K, and B is a member of K, then A + B is also in K.
(2) If A is a member of K, and B is a member of K, then A · B is also in K.
(3) There is an element 0 such that A + 0 = A for every A in K.
(4) There is an element 1 such that A · 1 = A for every A in K.

Laws:

(5)	$A + B = B + A$	commutative law of +
(6)	$A \cdot B = B + A$	commutative law of ·
(7)	$A \cdot (B + C) = A \cdot B + A \cdot C$	distributitive law
(8)	$(A + B) \cdot (A + C) = A + B \cdot C$	
	$(A + B) + C = A + (B + C)$	associative law (can be proved)

"Duality" - can interchange each + and · to get dual expression.

Inverses: (9) If 0 and 1 are unique, then there is an element $\overline{A}$
$A \cdot \overline{A} = 0$
$A + \overline{A} = 1$

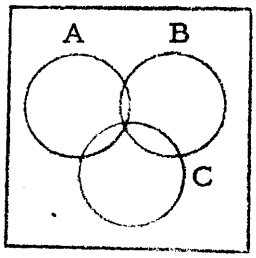
Also note:

$A + 1 = 1$	since	$1 + 1 = 1$	$0 + 1 = 1$
$A \cdot 0 = 0$		$1 \cdot 0 = 0$	$0 \cdot 0 = 0$

Notation: not standardized. The following are used in books:

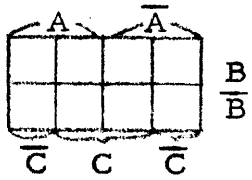
A and B	A or B	not A
$A \cdot B$	$A + B$	A, A^1, $\sim A$, A, $\neg A$
AB		
$A \cap B$	$A \cup B$	
$A \wedge B$	$A \vee B$	
$A + B$	$A \cdot B$, AB	

Venn Diagrams:

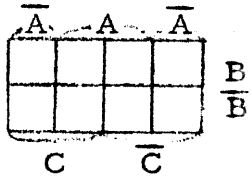


Show: $A + \overline{A}$
$A + B$
$A \cdot B$
$A + B \cdot C$
$(A + B) \cdot (A + C)$

Veitch Diagrams:



or



Truth Tables:

ABC	A	B, C	A+B· C	A+B	A+C	(A+B)· (A+C)
000	0	0	0	0	0	0
001	0	0	0	0	1	0
010	0	0	0	1	0	0
011	0	1	1	1	1	1
100	1	0	1	1	1	1
101	1	0	1	1	1	1
110	1	0	1	1	1	1
111	1	1	1	1	1	1

Note: $A + A = A$
$A \; A = A$

$A (B + \overline{B}) = A$

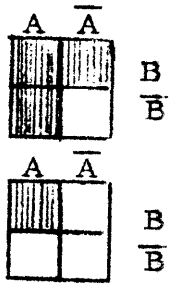
--- all big help in simplifying expressions

De Morgan's Theorems:

1. $\overline{(\overline{A})} = A$

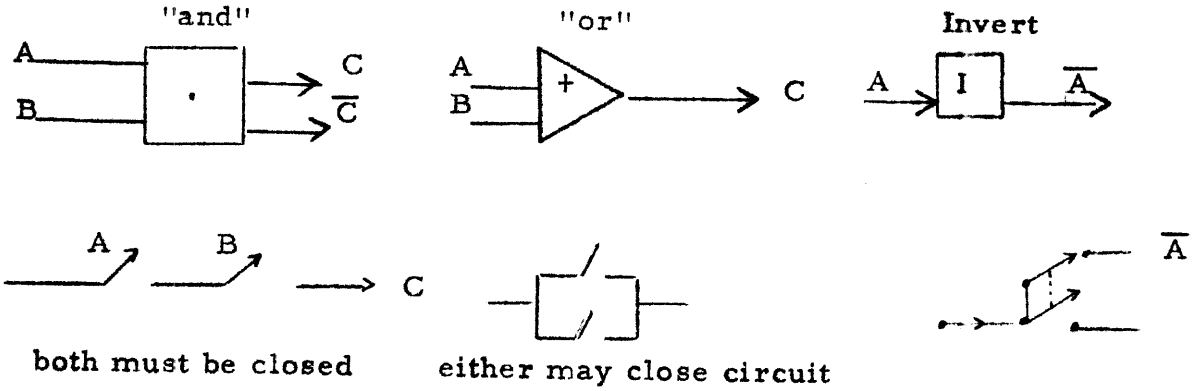
2. $\overline{(A + B)} = \overline{A} \cdot \overline{B}$

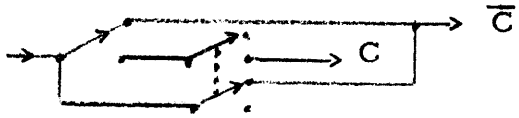
3. $\overline{A \cdot B} = \overline{A} + \overline{B}$



Rule: To complement a Boolean expression change all + signs to •, all • Signs to +, and replace each letter by its complement.

Circuit elements:

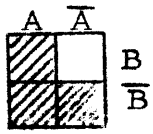




Boolean Functions:

functions can take many forms

$f = A + \overline{B}$



also $\overline{(\overline{A} \cdot B)}$ max term expansion

$= A + \overline{A}\,\overline{B}$

$= A\,B + \overline{B}$

$= A\,(B + \overline{B}) + \overline{B} = A\,B + A\,\overline{B} + \overline{B}$

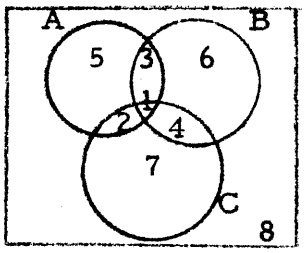
$= A + \overline{B}\,(A + \overline{A}) = A + A\,\overline{B} + \overline{A} + \overline{B}$

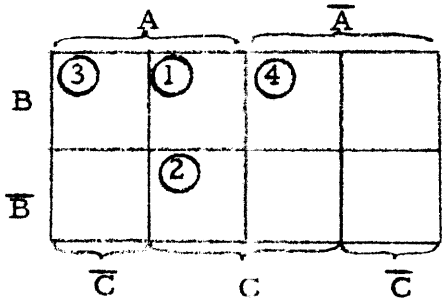
$= A\,(B + \overline{B}) + \overline{B}\,(A + \overline{A}) = A\,B + A\,\overline{B} + \overline{A}\,B$ ← The min term expansion

etc.

Problem 1: Draw the Truth tables for each of the six above representations of f.

Problem 2: Work out Vietch Diagram from the 3 variable Venn Diagrams.





Minterms & Maxterms & expansions.

The Basic Theorem: Any function of n variables can be expressed as a Boolean sum of a set of minterms.

The Minterm Expansion: $f = \sum_{i=0}^{2^n - 1} f_i\, m_i$ for n variables

Logical Connectives: All possible combinations of 2 input variables

	A	0	0	1	1	
	B	0	1	0	1	
Fo		0	0	0	0	F = 0 zero
1		0	0	0	1	= A B "and"
2		0	0	1	0	= $A\overline{B}$
3		0	0	1	1	= A
4		0	1	0	0	= $\overline{A}B$
5		0	1	0	1	= B
6		0	1	1	0	= $A\overline{B} + \overline{A}B$ "excl. or"
7		0	1	1	1	= $A + B$ "or"
8		1	0	0	0	= $\overline{A} \cdot \overline{B}$ "pierce" or $\overline{(A + B)}$
9		1	0	0	1	= $\overline{A}\,\overline{B} + A B$
10		1	0	1	0	= $\overline{B}$
11		1	0	1	1	= $A + \overline{B}$ inclusion
12		1	1	0	0	= $\overline{A}$
13		1	1	0	1	= $\overline{A} + B$
14		1	1	1	0	= $\overline{A} + \overline{B}$ Sheffer Stroke or $\overline{(A \cdot B)}$
15		1	1	1	1	= 1

Note: (9) $\overline{A}\,\overline{B} + A B = 1$ means A = B
(11) $A + \overline{B} = 1$ means B is included in A.

COMPUTER ORGANIZATION AND SYSTEMS

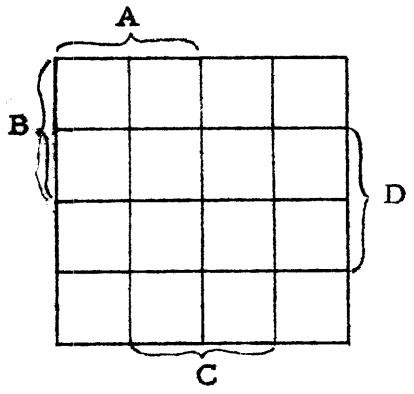
Lecture Notes

Lecture 7: Wednesday, April 25, 1962

Boolean Algebra (Continued)

Simplification of Boolean Expressions

1. "Cut and Try" method (Ref. Phister p. 66)(list simplifying terms)
2. Quine method
   Use of $XY + \overline{X}Y = Y$ on minterms
3. Nelson's method
   $A + AB = A$
4. Harvard Chart method
5. Veitch Diagram method (Ref. p. 84)



See p. 86,

$\overline{C}\overline{D} + AB\overline{D} + \overline{A}B\overline{C}$

Look for $XY + \overline{X}Y = Y$ Pattern

Use with "Don't Care" Cases:

In each term a letter appears as
0. Complimented form
1. True form
2. Doesn't appear - "don't care" case

There are 3^n-1 possible terms instead of 2^n-1.

Use with multiple functions -- (just mention)

The "Order" of Circuits: "Logical Levels"

1st Order $f = AB$, $f = A+B+C+D$

2nd Order $f = A+BC$

Output of one order = Input of Other
Delay in going through each -- may give "skew" in results.

Problem: Given the equations for a binary "half adder" :

$$S = A \cdot \overline{B} + \overline{A} \cdot B$$

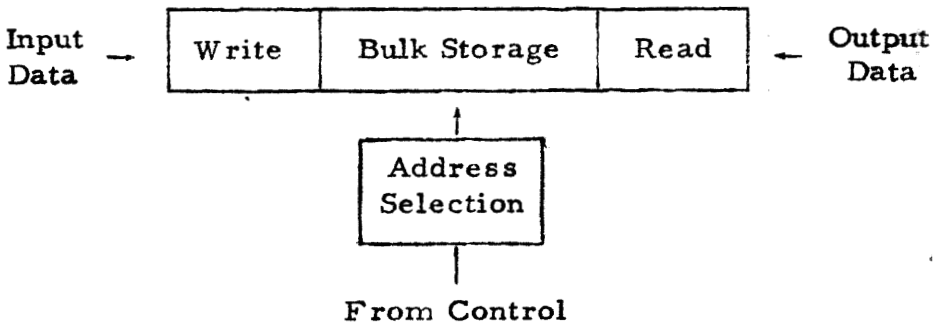
$$C = A \cdot B$$

(1) Write its truth table.

(2) Draw a circuit to give these results.

IV. Storage Devices

(Ref. Siegel - Chapt. 12)



3 Elements: (1) Storage, (2) Read and Write Circuits, (3) Address Selection

Characteristics: -- discuss each

1. Size: 1 bit - flip flop, triggers
   Word - register, counter, rings
   Multiword - bulk store -- core, drum, etc.

2. External - Internal -- define as "will the machine run without it?"

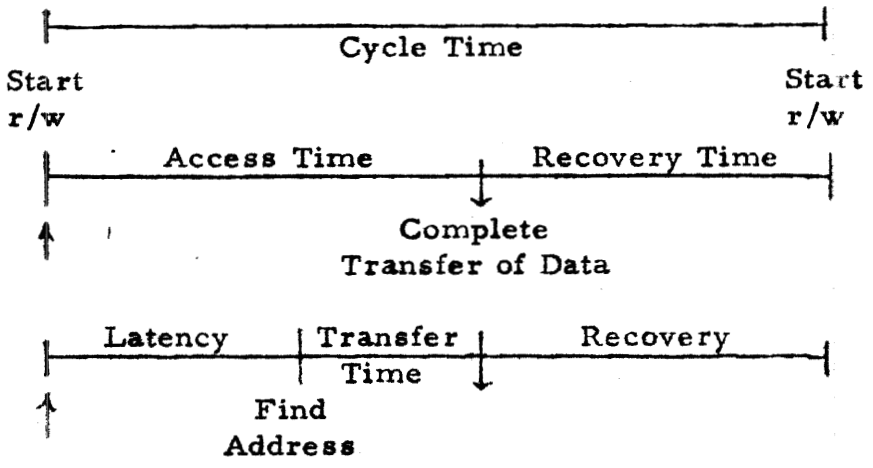
3. Fixed - erasable

   By "computer memory" we usually mean: bulk, internal, erasable.

4. Type of Access:

   Coordinate
   Cyclic - serial in time
   Sequential - parallel -- several bits at once

5. Latency, Access Time, Cycle Time



Note: People sometimes confuse "Transfer Time" with "Cycle Time".

6. "Random" Access

Define non-random access as having to wait differing amounts of time for data after giving read or write signal.

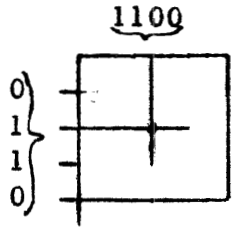
-- A relative measure depending on speed of storage and devices using it.

7. Address Selection: (Ref. p. 179, Phister) (Diagrams: Siegal p. 267)

$$f = \overline{S}_1 \cdot \overline{S}_2 \cdot \overline{S}_3 \cdot \overline{S}_4 \cdot \overline{S}_5 \cdot \overline{S}_6 \cdot \overline{S}_7 \cdot \overline{S}_8 \; P_0 + \text{etc.}$$

$$f = \sum_{i=0}^{255} (S_1 - S_8)_i \, P_i$$

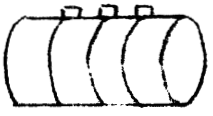
Spacial Addressing - Cores



Address: 01101100 Split into x and y address of core

Need $2(2^{n/2})$ instead of 2^n drivers.

Time Addressing - Drum



Time clock and track number
- count down to 0
- count up to address

8. Characteristics of Storage Elements Themselves

Static - dynamic (delay lines, etc.)
Erasable - non-erasable
Volatile - non-volatile (power off)
Destructive Read-Out - non-destructive

COMPUTER ORGANIZATION AND SYSTEMS

Lecture Notes

Lecture 8: Friday, April 27, 1962

1. Finish up discussion of Storage Devices
2. Central Processing Unit discussion (Assign read ahead on I/O)
3. Start Arithmetic
4. Movie, "IBM Teleprocessing"

[Problem: Write 1/2 page Summary]

IV. Storage Devices (Continued)

9. Cost - Cost per Bit

Ref. graph of cost per bit for cores, delay line revolvers, track revolvers, disks. (Give class copies and discuss).

10. Types of Storage Devices (Ref. Chapin Table P. 300)

(Give class copies and discuss each)

Mag tape
Mag drum
Mag disks
Mag cores
Punched cards
Paper tape
Acoustic delay lines
Electrostatic tubes
Photographic film
Vacuum tubes - transistors
Relays
Ferro-electric
Thin films - cryotrons

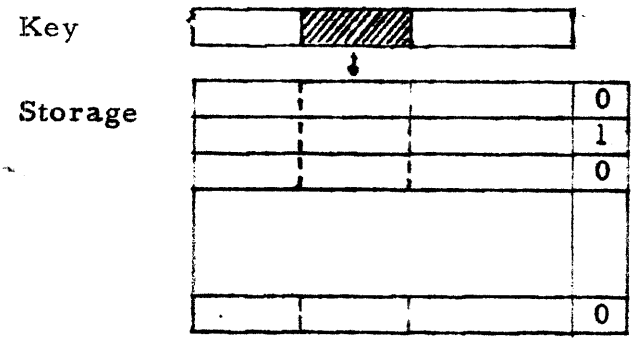
11. Future Developments

A. Read-only memories

Describe (a) photo-store
(b) capacitor memory
(c) SCAMP "Transformer" memory

B. Associative memories

Describe problem of finding match on key.



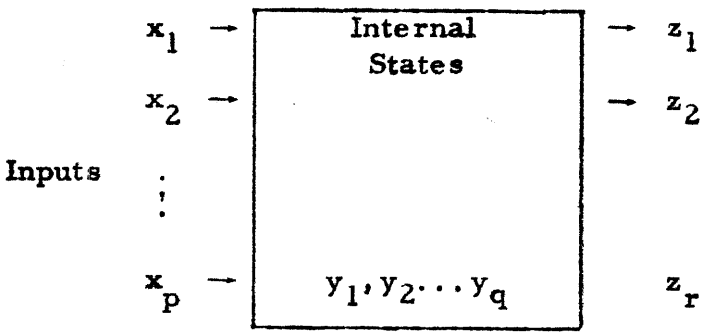
- Match on all words at once - mark those which do.
- Read out those which match.

C. Cryogenic memories

V. Central Processing Unit (Ref. Phister p. 144)

A. Data Flow

Huffman-Moore general model of a computer:

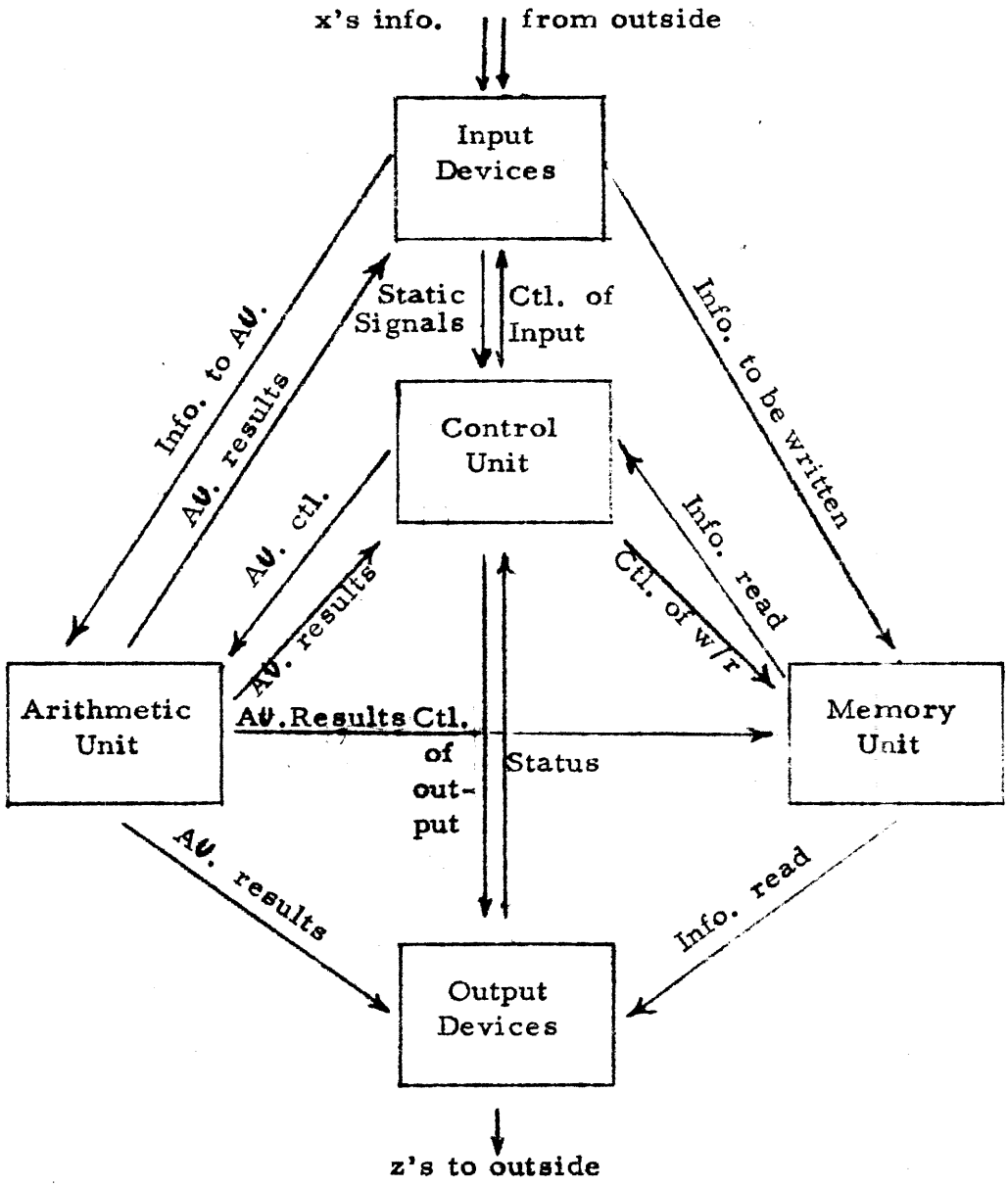


Define an "automaton".

At time n: $z_i^n = f_i (x_1, x_2, \cdots x_p, y_1, y_2, \cdots y_q)^n$

$$y_i^{n+1} = g_i (x_1, x_2, \cdots x_p, y_1, y_2, \cdots y_q)^n$$

Breakdown of Model:



B. Computer "Words" - Data Formats

Fixed word - examples of parallel fixed words.
Registers - length helps fix word length.
Data "Buses" - also fixed by word length.

H. G. Kolsky

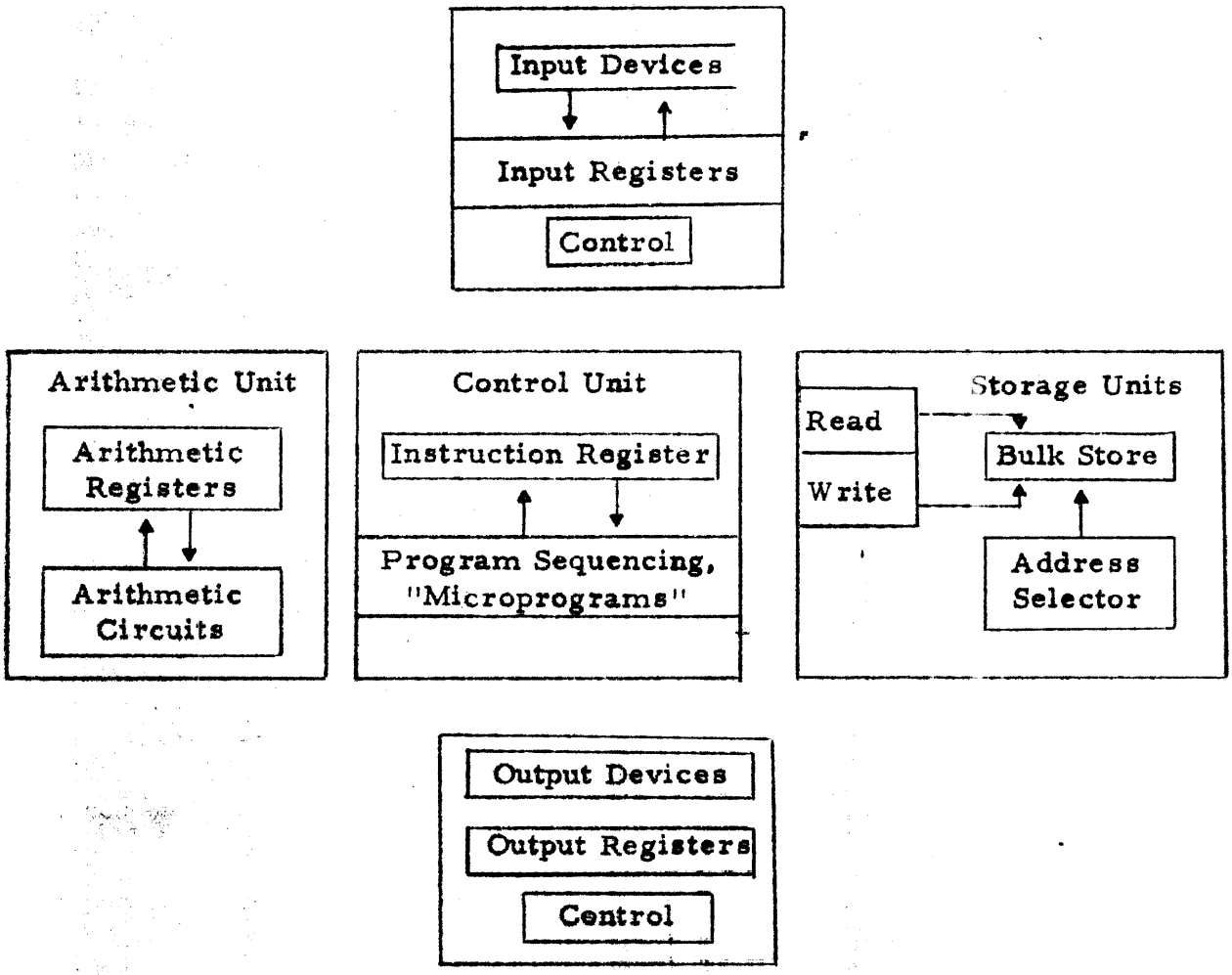
COMPUTER ORGANIZATION AND SYSTEMS

Lecture Notes

Lecture 9: Monday, April 30, 1962

V. Central Processing Unit (cont)

Draw general Circuit Diagram -- discussing as we go --



C. Control Circuits

Concepts:

word length
registers -- (discuss above registers)
buses

Mention use of instructions as ordinary data . . . same storage, same rules of modification. (important concept. Attributed to von Neuman)

Instruction Formats:

Instruction Register (or location counter)

Location Counter: Operating Code Address(es)

Discuss:
(a) 4 address L:A op B = C Go to D
(b) 3 address L:A op B = C (Go to next instr.) - Need "Branches"
(c) 2 address L:A op B = (acc) (Go to next instr.) - Need "Stores"
(d) 1 address L:A op (acc) = (acc) (Go to next instr.) - Need "Loads"

Go through an example programs: (If there is time)

$F = (A + BX)/Dy$

4 Address Program

I : (D) Mpy (y) put in (T_1) go to (I + 1)
I+1: (B) Mpy (x) put in (T_2) go to (I + 2)
I+2: (A) Add (T_2) put in (T_2) go to (I + 3)
I+3: (T_2) Div. by (T_1) put in (F) go to (next prog.)

3 Address Program

Same as above with no "go to", plus

I+4: Branch to (next prog.)

2 Address Program

I : (D) Mpy (y)
I+1: (acc) Store (T_1)
I+2: (B) Mpy (X)
I+3: (acc) Add (A)
I+4: (acc) Div. by (T_1)
I+5: (acc) Store in F
I+6: Branch to (next prog.)

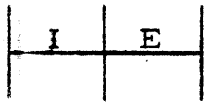
1 Address Program

I : Load (D)
I+1: Mpy (y)
I+2: Store (T_1)
I+3: Load (B)
I+4: Mpy (X)
I+5: Add (A)
I+6: Div. by (T_1)
I+7: Store (F)
I+8: Branch to (next prog.)

Discuss "bit efficiency" and simplicity of decoding.

Machine Cycles:

(1) Instruction Cycle (I-cycle)
Instruction fetch using Loc. Ctr.
Decode operation
Operand data address found
Location counter is set to next instr.



(2) Execution Cycle (E-cycle)
Operand fetched
Arithmetic operation performed
Result returned to accumulator

Machine sub-cycles
Describe micro-programming briefly.

Interruption and testing
Overlapping

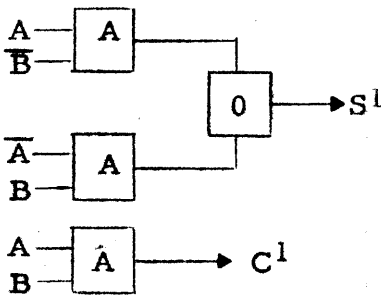
D. Serial and Parallel Arithmetic

(1) Addition:

Binary half adder: 2 input - 2 output

$S^1 = A\,\overline{B} + \overline{A}\,B = A \vee B$
$C^1 = A\,B$

A	B	C^1	S^1
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



S^1:

	A	$\overline{A}$
B	0	1
$\overline{B}$	1	0

C^1:

	A	$\overline{A}$
B	1	0
$\overline{B}$	0	0

Binary Full Adder: 3 input - 2 output

A	B	C	S	Co
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Veitch Diagrams:

S:

	A	$\overline{A}$	A	$\overline{A}$
B	1	0	0	1
$\overline{B}$	0	1	1	0
	C		$\overline{C}$	

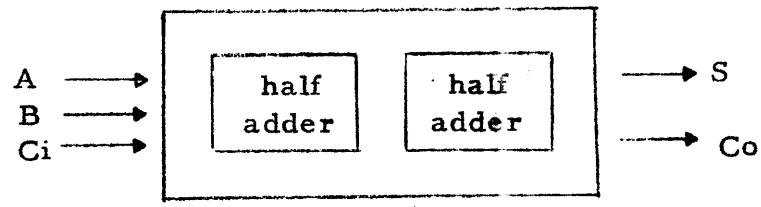
C:

	A	$\overline{A}$	A	$\overline{A}$
B	1	1	1	0
$\overline{B}$	1	0	0	0
	C		$\overline{C}$	

Derive Equations from Table:

$$S = \overline{A}\,\overline{B}\,C + \overline{A}\,B\,\overline{C} + A\,\overline{B}\,\overline{C} + A\,B\,C$$
$$Co = \overline{A}\,B\,C + A\,\overline{B}\,C + A\,B\,\overline{C} + A\,B\,C$$

Problem: Given these equations show how one can get effect of full adder from two half adders.



H. G. Kolsky

COMPUTER ORGANIZATION AND SYSTEMS

Lecture Notes

Lecture 10: Wednesday, May 2, 1962

Solution of Problem:

$$S = (\bar{A}B + A\bar{B})\,\bar{C} + (\bar{A}\bar{B} + AB)\,C$$

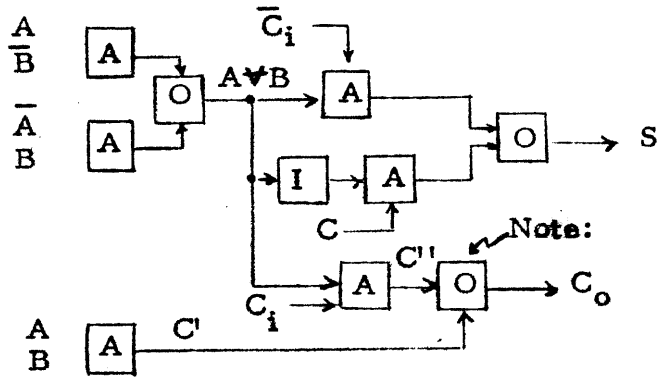
$$= (\bar{A}B + A\bar{B})\,\overline{C} + \overline{(AB + A\bar{B})}\,C$$

$$= S'\bar{C} + \bar{S}'C$$

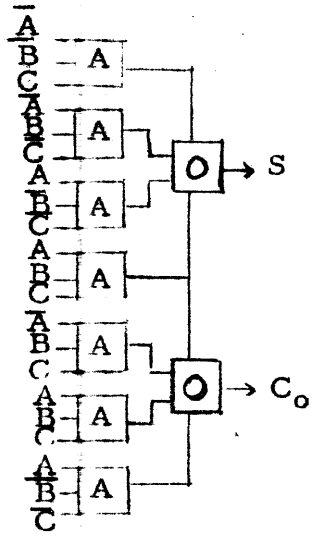
$$Co = (\bar{A}B + A\bar{B})\,C + AB(\bar{C} + C)$$

$$= S'C + C'$$

Circuits - Full Adder

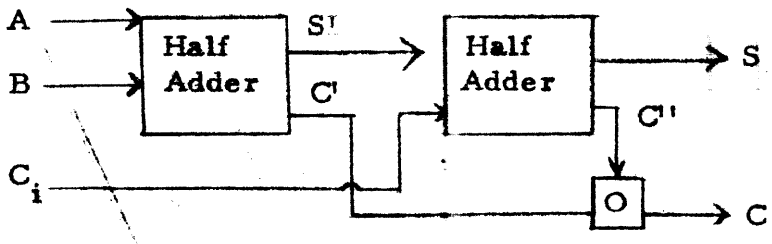


2 half-adders



1st half

A	B	C_i	S'	C'	S	C''	C_o
0	0	0	0	0	0	0	0
0	0	1	0	0	1	0	0
0	1	0	1	0	1	0	0
0	1	1	1	0	0	1	1
1	0	0	1	0	1	0	0
1	0	1	0	1	0	1	1
1	1	0	0	1	0	0	1
1	1	1	0	1	1	0	1



(2) Subtraction:

(Go through only if there is time)

Half Subtractor:

A - B	
A	B
0	0
0	1
1	0
1	1

S' + B'	
B'	B'
0	0
1	1
0	1
0	0

Full Subtractor:

A	-	B	B_i	B_o	S
0		0	0	0	0
0		0	1	1	1
0		1	0	1	1
0		1	1	1	0
1		0	0	0	1
1		0	1	0	0
1		1	0	0	0
1		1	1	1	1

Subtraction by complements:

```
 1 0 1 1 0 1 0
-0 1 0 1 1 0 0
---------------
 0 1 0 1 1 1 0
```

```
 1 0 1 1 0 1 0
 1 0 1 0 0 1 1   1's complement
             1   +1
1|0 1 0 1 1 1 0
```

Explain need for 1 in -A = C(A) + 1

(i. e., +0 and -1)

Discuss number of carries in a parallel additon:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

1 out of 4 on first half-add

```
                    1 0 1 1 1 1 0
                    0 1 1 0 1 0 0
                    ---------------
1st cycle S':       1 1 0 1 0 1 0
          C':       0 0 1 0 1 0 0
                    ---------------
2nd cycle S'':      1 0 0 0 0 1 0
          C'':      0 1 0 1 0 0 0
                    ---------------
3rd cycle S''':     0 0 1 0 0 1 0
          C''':     1 0 0 0 0 0 0
                    ---------------
4th       S^IV: 1   0 0 1 0 0 1 0
          C^IV:     0 0 0 0 0 0 0
```

Expected number of carry cycles ?

1's first cycle = $\frac{N}{4}$

1's 2nd cycle = $\frac{N}{4} \cdot \frac{1}{2}$

3rd cycle = $\frac{N}{4} \cdot \frac{1}{2} \cdot \frac{1}{2}$

Worst case:

```
1 1 1 1 1 1
0 0 0 0 0 1
```

Best case:

```
1 0 0 1 0
0 1 1 0 1
```

$$\frac{N}{2^{n+1}} < \frac{1}{2}$$

So for 32 bits we would expect ~ 5 carry cycles, not 32.

--Mention skipping of 1's:

Worst case becomes

```
1 0 1 0 1 1 1
1 0 0 0 0 0 1
```

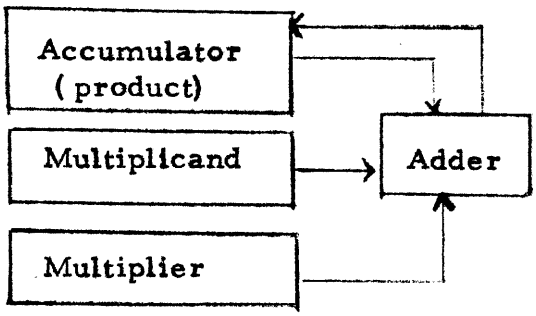
(3) Multiply:

A x B = C

Length of product field = length A + length B

How to handle double length ?

Mention scaling



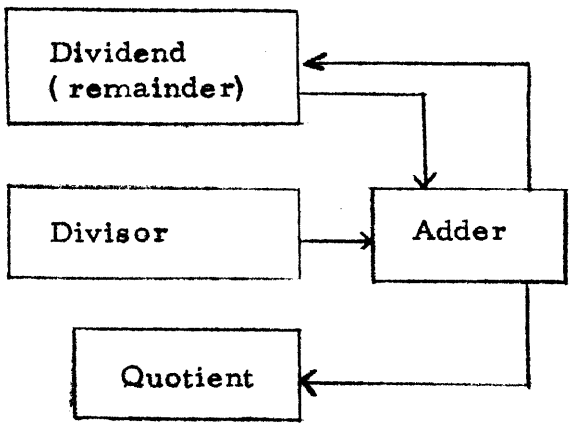
Multiplier controls number of times Multiplicand is added in.
Multiplicand register is usually the memory register.

Example:

```
                    1 0 1 1
                    1 1 0 1
Acc. :              0 0 0 0      Shift and add or no-add
(1)                 1 0 1 1  <---Keep sum as we go along
                    1 0 1 1
(2)                 No add
                0 0 1 0 1 1
(3)               1 0 1 1        N^2 add cycles
              0 1 1 0 1 1 1
(4)           1 0 1 1            Double length answer
          1 0 0 0 1 1 1 1        (H.O.) + (L. O.) product
```

(4) Divide: $A/B = Q + \frac{R}{B}$ Two answers Q + R

Length of quotient = length of A - length of B



Quotient is generated by repeated subtraction until "over-drawn"

Do example if time:

```
                    1 0 1 1.    = Q
              ┌──────────────
1 1 0 1   /   1 0 0 1 0 0 0 0.
(C + 1)   1 0 0 1 1
          ─────────
            1 0 1 0 0
            1 0 0 1 1
            ─────────
              0 1 1 1 0
              1 0 0 1 1
              ─────────
                0 0 0 1   = R
```

(5) Square Root

Use algorithm $N = (a + b)^2$

$= a^2 + 2ab + b^2$

Remainder $= N - A^2 = (2a + b)\, b$

Give more if there is time.

(6) Logical

Just mention--covered earlier

(7) Branching:

I. Conditions

a. Unconditional
b. Conditional on counts
c. Conditional on arithmetic results
d. Conditional on interrupt

II. Method of Branching

a. In 4 address machine--have two addresses (1) and (2)
b. In 3 or less, (1) go to next instr., (2) other address given
c. Skip an instr. (give example of CAS)
d. Indirect--get next instr. from table

Mention how branch may be on the condition being satisfied or not being satisfied.

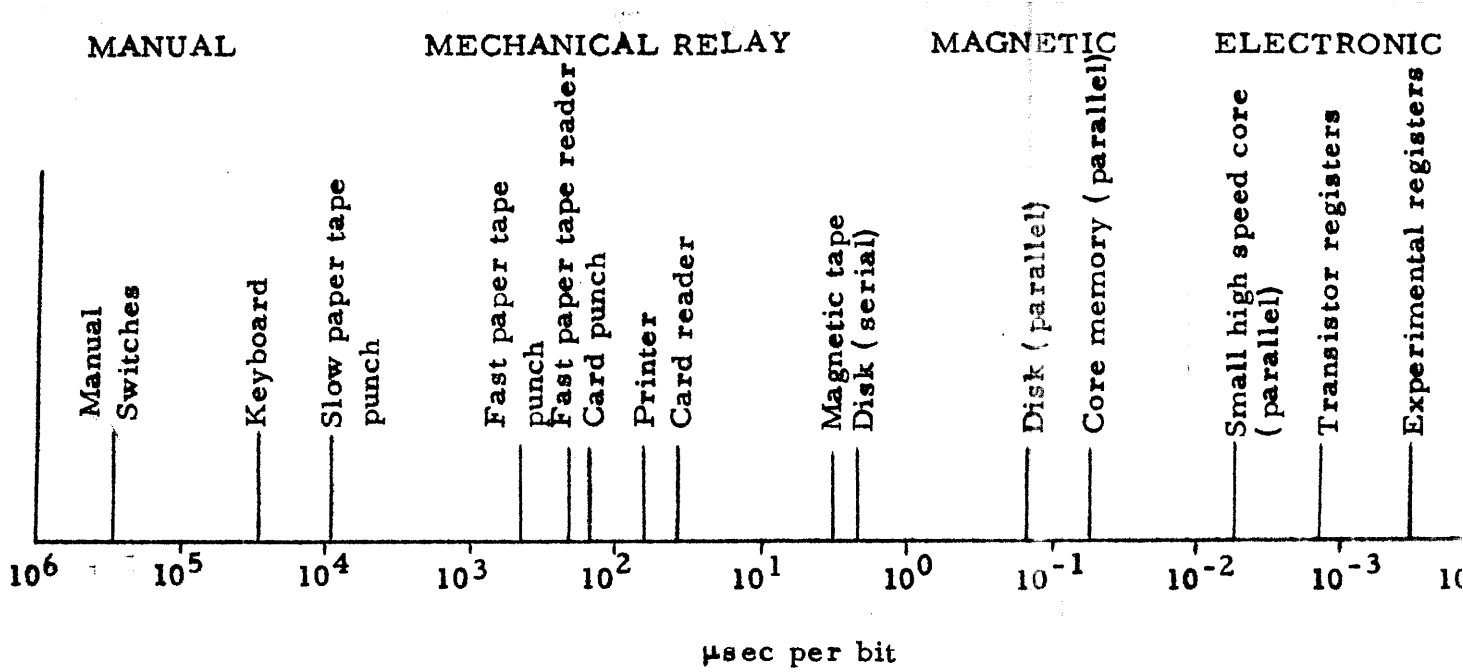
May 4, 1962

Lecture 11

VI. Input-Output Devices

A. Speed Hierarchy

Device	Max. Bit Rate	Time Per Bit
1. Manual Switching (3 per sec)	3. bits / sec	333, 000. μsec/ bit
2. Keyboard: (5 char/ sec)	30	33, 000.
3. Slow paper tape punch (15 char. per sec)	90.	11, 000.
4. Fast paper tape punch (300 char/ sec)	1, 800.	555.
5. Paper tape reader (500 char/ sec)	3, 000.	333.
6. Card punch (250 cds/ min)	4, 000.	250.
7. Printer (1000 l/ min)	12, 000.	83.
8. Card reader (1000 cds/ min)	16, 000.	62. 5
9. Magnetic tape (62. 5 KC)	375, 000.	2. 7
10. Disk 1301 (125 KC)	750, 000.	1. 33
11. Disk (STRETCH) (64 bits in 8 μsec)	8, 000, 000.	0. 125
12. Core memory (36 bits in 2. 2 μsec)	16, 400, 000	0. 061
13. Small core register (64 bits in 0. 4 μsec)	160, 000, 000.	0. 006
14. Transistor registers (64 bits in 0. 1 μsec)	640, 000, 000.	0. 0016
15. Experimental circuits (~ 5 μsec)	~ 5, 000, 000, 000.	0. 0002



B. Devices:

Card readers (ref. p. 43 of text)
Card punches
Magnetic tape units
Paper tape reader
Paper tape punch
Printers
- bar
- wheel
- wire
- chain
- typewriter

CRT
Consoles

Discuss above if there is time.

FINAL EXAM. Second Hour

ENGINEERING TRAINING PROGRAM

FINAL EXAMINATION

May 25, 1962

COURSE: 7090 System

INSTRUCTOR: H. G. Kolsky

I. Underline the correct word within each set of parentheses:

The 7090 is a (binary, decimal) computer utilizing (serial, parallel) arithmetic. The first 7090 was completed in (1961, 1959, 1957). It has (5, 3, 9) index registers which are used for (address, storage) modification. It has a storage capacity of (16, 384; 4096; 32, 768; 1025) words of (36, 48, 72) bits each. The 7090 can execute nearly (200, 000; 50, 000; 500, 000; 2, 000, 000) instructions per second. It can have a maximum of (2, 4, 8, 16) data channels, each of which can hold up to (2, 6, 8, 10) tape units.

II. Draw a diagram of the main registers of the 7090 and the data paths between them. (Be sure to include the position of the adder.)

III. Draw the following formats giving the bit numbers for the start and end of each field:

(1) A Fixed Point Word



(2) A Floating Point Data Word



(3) An Index-Type Instruction Containing Address, Prefix, and Tag.



*IV. Write a machine language program to calculate Sin F. Single precision (floating or fixed), about 8 decimals accuracy. Assume F is in storage at loc F. Assume Sin F is to be put in loc F + 1. Refer to pages 96-99 of the manual for notation.

*(Problem IV is to have been done as homework before-hand and handed in with the final.

H. G. Kolsky
May 4, 1962

FINAL EXAM

COMPUTER ORGANIZATION & SYSTEMS

A. Define or explain each of the following terms as used in the course:

1. Bit

2. Code

3. Truth Table

4. Redundancy

5. System

6. System constraint

7. Veitch diagram

8. half adder

9. logical connectives

10. weighted code

B. Answer each of the following questions briefly.

1. What computer is usually named as the "trunk" of the family tree of computers? What date was it completed?

2. What are the three main "branches" of the computer family tree?

3. Name two methods of converting numbers from one base to another.

4. Name the five main subdivisions of a computer.

5. Which topic covered in the course do you feel was the most useful or most interesting and why?

H. G. Kolsky
May 1962

REFERENCES ON COMPUTERS IN RESEARCH-ASDD LIBRARY

"Origin of Modern Calculating Machines", J. A. V. Turck (1921)	QA75 T932
"Punched Card Methods in Scientific Computation", W. J. Eckert (1940)	QA75 E3
"Manual of Operation (Harvard Mark I)" (1946) Interesting early machine	QA75 H325
"Electronic Digital Computers", J. W. Forrester (1947)	MIT R-116
"Computing Mechanisms and Linkages", A. Svoboda (1948)	QA75 S85
"Proc. of Second Symposium on Large Scale Digital Calc. Machinery" (1949) Very good early work	QA76 S9
"Giant Brains or Machines That Think", E. C. Berkeley (1949)	QA75 B4
"Review of Electronic Digital Computers", AIEE Conf. (1951)	QA76 A55
"Computing Manual", F. J. Gruenberger (1952)	QA75 G775
"Proceedings of the ACM" (May, 1952) Good history and early papers	QA76 A89
"Punch-Card Methods", H. P. Hartkemeier (1952) Wiring of various machines, incl. 101 and 407	
"Faster than Thought", B. V. Bowden (1953) Good history	QA76 B68
"Survey of Domestic Electronic Digital Computing Systems", U. S. Dept. of Commerce (1955)	QA76 W421
"Arithmetic Operations in Digital Computers", R. K. Richards (1955)	QA75 R5
"An Introduction to Automatic Computers", Ned Chapin (1955)	HF 5548 C463
"Computers, their Operation and Application", E. C. Berkeley (1956)	QA76 B4
"Electronic Data Processing for Business and Industry", R. G. Canning (1956)	HF 5548 C245

"Automatic Digital Computers", M. V. Wilkes (1956) EDVAC, EDSAC, Babbidge	QA76 W55
"Automatic Digital Calculators", A. D. Booth (1956)	QA76 B6
"Electronic Digital Computers", F. L. Alt (1958)	QA76.5 A65
"Switching Circuits and Logical Design", Caldwell (1958)	TK2821 C147
"IBM Engineering Facilities Conference on Digital Computers" (1958) (Also contains bibliography)	QA75 I610
"The Computer and the Brain", J. Von Neuman (1958)	QA76 V945
"The ABC's of IBM Machine Operation", L. J. Salmon (1959) Chapters on each of main types of EAM equipment including sorters, calculators, interpreters, reproducers, collators, etc.	HF5520 S172
"Today's Business Machines", H. Fisher (1959)	HF5548 F533
"Computers and Data Processing", E. M. Grabbe (Handbook) (1959)	TJ213 G727
"Office Automation - Operations Research", R. H. Brown (1959) Survey of equipment	QA76 B879
"Symbolic Logic and Intelligent Machines", E. C. Berkeley (1959)	QA76 B512
"Association of Computing Machinery, 1959 Annual Meeting	QA76 A849
"Computer Engineering", S. A. Lebedev (1960) (Translation from Russian)	QA76 A313
"Computer Logic", I. Flores (1960)	QA76.5 F634
"Digital Computers and Control Engineering", Ledley (1960) Good on timing	QA76.5 L473
"Electronic Computers, Principles and Applications", T. E. Ivall (1960)	TK7885 I930
"Revised Specifications for a Common Business Oriented Language", CDSL (COBOL 1961)	HF5548 C748

"Computer Programming Fundamentals", A. D. Leeds (1961) QA76.5 L484

"A Third Survey of Domestic Electronic Digital Computing Systems", Department of Commerce PB-171265 (1961) QA76 W421
Good Survey

"Understanding Digital Computers", Paul Siegel (1961) QA76.5 S571

"Computing for Scientists and Engineers", R. W. Hamming (1961) QC76.5 H224

"Electronic Data Processing", E. W. Martin (1961) HF 5548 M379

"Electronic Data Processing and Auditing", F. Kaufman (1961) HF5679 K210

"Electronic Digital Computers", G. D. Smirnov (1961) QA76.5 S641
(Translated from Russian)

"Advances in Computers", edited by F. L. Alt (1961) QA76 A244
Good history (mostly applications)
Read

GPD Tech. Record	Digital Computers - Logical Principles and Operation	TD 01. 02. 003. 005 (1960)
IBM (S)	Data Processing Bibliography	Form J20-8014-1 (1959)
IBM (S)	Introduction to IBM Data Processing Systems	F22-6517-0 (1960)
IBM (S)	Planning for an IBM Data Processing System	F20-6088-1 (1961)
IBM (S)	Advanced Analysis Method for Integrated Electronic Data Processing	F20-8047 (1960)
IBM (S)	Document and Accounting Controls	C20-8060 (1961)
IBM (S)	Index Organization for Information Retrieval	C20-8062 (1961)
IBM (S)	Introduction to Engineering Analysis for Computers	F20-8077 (1961)
IBM (M)	Operator's Guide - Unit Record Equipment	A24-1010 (1959)
IBM (M)	Functional Wiring Principles	A24-1007 (1960)

MACHINE TYPES

1. Punches verifiers
2. Sorters
3. Accounting machines
4. Calculators
5. Auxiliary machines (Incl. 101, 357, 077, 1001, etc.)
20. Proof machines - test scoring
22. 1620 system
25. 1200 system
27. 1401 system
28. 1410 system
30. 305 RAMAC
40. 650 systems
50. 700 and 7000 systems
60. 700 and 7000 Auxiliary machines (Printing tapes, etc.
70. Control systems (1710)
90. Information Retrieval
99. Miscellaneous

IBM (M) Machine Load Computer, Form X24-8704-1 (Get one)

Movies possible?

F-15.3	IBM (F)	"The Information Machine" Color 16mm, 10 min.	(N.G.)
F-15.6	IBM (F)	"IBM Teleprocessing" Black and White, 16mm, 13 min.	(O.K.)
F-15.20	IBM(F)	"Principles of Electronic Data Processing" Color 16mm, 16 min.	
F-15.31	IBM (F)	"Sabre System" Color 16mm, 17-1/2 min.	

Filmstrips on 1001, 1401, 1410, 1620
JobShop Simulation, Princ. of EDP

COMPUTER ORGANIZATION AND SYSTEMS
April 9, 1962
CLASS ROLL

NAME	AGE	MAN No.	DEPT. No.	DEPT. NAME	SCHOOL	Degree, Subject Year	PRESENT JOB	PREVIOUS JOB EXPERIENCE
S. Azar	29	947044	477	Control Systems D. L.	Univ. of Mich.	MS E. E. 1959	Design & Develop., Advanced Computer Contr. Systems, Solid-State Circuitry.	Design: Aircraft Electrical Systems
Frank H. Brittain	23	211362	492	Mechanical Analysis Lab.	Univ. of Illinois Univ. of Colorado	Theo. & Appl. Mech., 1961 M. E. 1960	Mechanical analysis of various machines to help improve reliability and performance.	Univ. of Ill. - Research Asst., Non-linear mech. Lawrence Rad. Lab. - environmental testing. Collins Radio Co. Mech. Design.
Yih Ming Chu	26	516601	492	Mechanical Analysis Lab.	Natl. Taiwan Univ. Univ. of UTAH	BS M. E. 1956 MS M. E. 1962	Design engineering	Design work
Leslie G. Holmes	27	644623	547	Machine Technology - Circuits Review	Long Beach State Coll.	BS E. E. 1/1962	Circuit Review of 609 Circuits	Automatics: Systems on Autocheckout ept. Nortronics: Systems tech. on APAC(Airborne Parabolic Arc comp) Brain of Snark Missile.

NAME	Age	MAN No.	Dept. No.	DEPT. NAME	SCHOOL	Degree, Subject, Year	PRESENT JOB	PREVIOUS JOB EXPERIENCE
Max W. Kuypers	39	854014	525	Control, Spec. Systems	Royal Tech. College	Comm. Eng. 1943	Systems Engineering	Special systems design
Gim H. Leong	27	856331	567	MARS	Univ. of Wash. Col. City of New York	BS E. E. 1962 BS M. E. 1958	Control Logic	Electronic Circuit Analysis & Calibration, Hydraulic Systems, Mechanisms & Structures.
Samuel H. Odom	27	070582	585	Special Systems Engr.	Univ. of Texas	BS. E. E. 1961	Design implementation & release of spec. features to be installed in a customer's system.	Processing systems (IBM-S. J.) Technician (Mfg. Scintillation eqpt. Sanarmon, U. S. N.)
J. L. Overacker	27	671450	526	Mech. Design	San Jose State	BS ME 1961	Design of machine cooling systems, Structural analysis & mach. environ. control	
Dieter P. Paris	34	572281	522	Optics Techn.	Univ. of Hamburg & Erlangen	PHD. Physics 1958	Basic research in non-linear optical image formation	2 1/2 yrs. -Research in Optics & Photog. (ft. Monmouth, NJ) also: 1/2 yr. at Boeing, Seattle.
S. Razi	32	076183	477	Control Sys., Advanced Dev.	Mass. Inst. Tech.	MS EE 1953		3 1/2 yrs: Link Aviation 1 yr.: Burroughs Cor. 9 mos.: Photon, Inc.
Jack Y. Robertson	30	678251	479	Control Systems	Univ. of Calif.	BS EE, 1957 MS EE, 1959	Circuit design, in Analog to digital converter area.	Design of high-speed ferrite core memory.

CLASS ROLL (cont'd.)

NAME	Age	MAN No.	Dept. No.	DEPT. NAME	SCHOOL	Degree Subject, Year	PRESENT JOB	PREVIOUS JOB EXPERIENCE
Donald M. Sorensen	26	785571	549	Human Factors Eng.	San Jose State	BA Psy. 1959	Automated Teaching (Rand D)	none
W. S. Wu	31	499074	472	Control System	Mass. Inst. Tech.	Sc D. ME 1959	Computer Control Appl. Dev.	Heat Transfer, combustion, high-speed fluid mech. Control theory & instrumentations
Wen Cho Yang	29	599241	546	Package Group	Natl. Taiwan Univ. Univ. of Utah	BS ME 1956 MS ME 1962	Design engineer	design engineer